

POLY PMNA

REVISION 2026

COURSE 1182

PRACTICUM

Fundamentals of Architecture

A complete design-and-visual-language handbook covering elements of design, colour theory, texture, principles of design, freehand sketching and the architectural features of celebrated Indian and international buildings.

Course code	1182
Semester	I
Type	Practicum
Scheme	3:1:2:0
Contact hours	90
Credits	5
Summary CIE	40
Summary ESE	100

Programmes: Architecture and Interior Design.

OFFICIAL SOURCE DECLARATION

Official Source Declaration

This handbook is an original educational expansion of the supplied SITTTT Kerala Diploma Curriculum Revision 2026 syllabus PDF for Course 1182. Content is derived from the official syllabus, expanded with bilingual explanations, practical records, diagram exercises, worked examples, examination preparation and self-learning activities. No model-paper text is reproduced. Where syllabus wording is quoted, it is attributed.

All learning outcomes, module titles, contact-hour allocations and assessment weights follow the official syllabus document.

LEARNING ROADMAP

Course map and module overview

This practicum handbook develops the fundamental visual and representational skills of architectural design. It begins with the primary elements of design, progresses through colour theory and texture, then the principles that organise those elements, and culminates in the study and sketching of famous Indian and international buildings.

Module I

Elements of Design

Theory 12 h Practical 8 h

Module II

Colour and Texture

Theory 12 h Practical 6 h

Module III

Principles of Design

Theory 12 h Practical 10 h

Module IV

Famous Buildings

Theory 18 h Practical 12 h

OFFICIAL OBJECTIVES

Course objectives and outcomes

The official objectives are reproduced verbatim from the SITTR Kerala syllabus PDF.

Course objectives

1. To enable the students to identify the basic elements and principles of architecture design.
2. To introduce the fundamental techniques of visual design through drawing, painting and drafting.
3. To comprehend the basics of free hand sketching appropriate to visual and architectural representation.
4. To comprehend the concepts and architectural features of buildings designed by well known Indian and international architects.

Course outcomes

CO1

Apply fundamental architectural drawing skills and design principles

Represent elements of design concepts through clear, accurate, and well-composed drawings.

Bloom: Apply

CO2

Apply colour theory and texture concepts

Apply colour theory, colour schemes, colour psychology, and texture concepts to create effective visual compositions using appropriate materials and design principles.

Bloom: Apply

CO3

Apply basic design principles

Apply basic design principles — balance, proportion and scale, rhythm, hierarchy, contrast, harmony, and emphasis — by developing design compositions and outdoor sketches that effectively represent form, depth, lighting, and spatial quality.

Bloom: Apply

CO4

Apply design concepts of famous buildings

Apply the design concepts and architectural features of selected Indian and international buildings and represent their form, proportions, and spatial composition through freehand sketches.

Bloom: Apply

MODULE I • CO1**Elements of Design**

Theory 12 h

Practical 8 h

CO1

1.1 Basics of architectural drawing (3 h · Understand)

Architectural drawing is the universal language through which ideas, spatial relationships and construction intent are communicated between architect, engineer and builder. It begins with the disciplined layout of the drawing sheet: selecting the appropriate size, establishing a clean title block with course name, drawing title, date, scale and student identification, and arranging marginal guidelines. Line weight is a deliberate hierarchy — thick lines define the visible outline, medium lines show hidden or secondary features, and thin lines carry dimensions, text and centre-lines. Lettering must be uniform in height, spacing and slant; Roman uppercase is the international standard for titles. Scale converts real dimensions to paper size; common architectural scales include 1:100 for plans, 1:50 for sections and details, and 1:20 for enlarged details. Every drawing must carry a scale bar so that measurements can be verified even after reproduction.

Sheet layout includes border, title block and drawing area with consistent margins

Line weight hierarchy: thick for visible edges, medium for secondary, thin for dimension and construction lines

Lettering: uniform height, single-stroke Roman uppercase for titles and notes

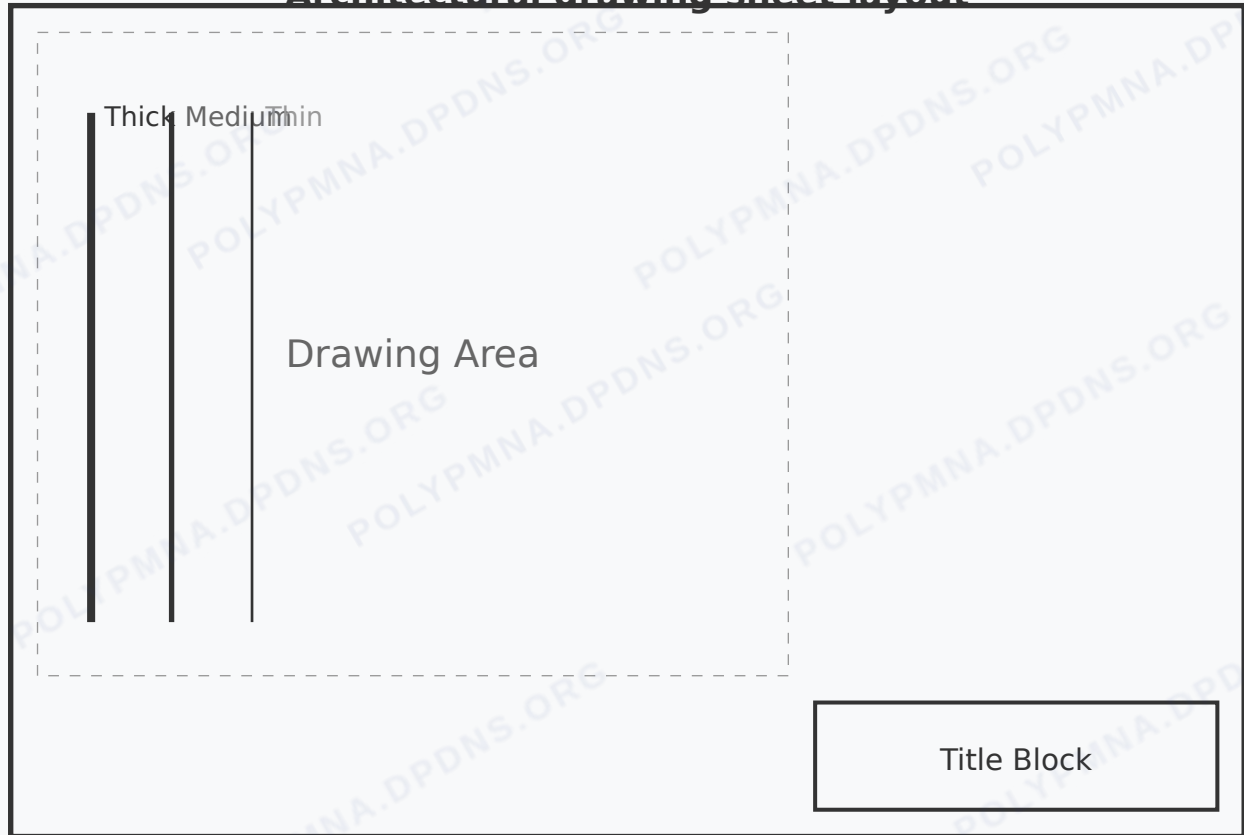
Scale: ratio of drawing dimension to real dimension; always show scale bar

Three standard architectural scales: 1:100 plans, 1:50 sections, 1:20 details

Every drawing must have title, date, scale, name and index number

Malayalam support: Sheet layout-ൽ border, title block, drawing area ഉണ്ട്. Line weight thick, medium, thin എന്നിങ്ങനെ hierarchy ഉണ്ട്. Lettering uniform ആയിരിക്കണം. Scale ratio drawing dimension : real dimension ആണ്.

Architectural drawing sheet layout



A3 Sheet · Border 10mm · Title Block 60×20mm

Architectural drawing sheet layout. Identify parts, trace visual/spatial flow, then state one application and one design significance.

Architecture connection

Relate this concept to an actual building, spatial experience, material choice or cultural context. Clearly state the design intent, visual strategy and environmental response.

Self-check

Define it, identify or draw its main features, explain the design principle, compare one alternative and state an architectural application plus limitation.

1.2 Elements of design: point, line, plane and volume 6 h · Understand

The four primary elements — point, line, plane and volume — are the building blocks of every architectural composition. A point has position but no measurable dimension; it marks a location, anchors attention, or suggests a node in a spatial sequence. A line is the path of a moving point; it carries direction, length and character. Vertical lines convey stability and aspiration; horizontal lines suggest rest and extension; diagonal lines create dynamism and tension; curved lines introduce softness and organic rhythm. A plane is the extension of a line in a second dimension — it defines surfaces, enclosures and boundaries. Volume extends the plane into three dimensions, giving the composition mass, weight and spatial presence. Form is the combination of visual properties (shape, size, colour, texture) and relational properties (position, orientation, visual inertia). Forms may be regular (geometric, predictable, composed of primary shapes) or irregular (asymmetric, organic, composed of secondary shapes).

Point: position only, marks location, creates focal emphasis

Line: path of a point; types include vertical, horizontal, diagonal, curved, zigzag, spiral

Plane: two-dimensional surface, defines boundaries and enclosures

Volume: three-dimensional extension, gives mass and spatial presence

Regular forms: geometric, predictable (cube, sphere, cylinder, cone, pyramid)

Irregular forms: organic, asymmetric, complex composites

Form properties: visual (shape, size, colour, texture) and relational (position, orientation, inertia)

Malayalam support: Design-ന്റെ അടിസ്ഥാന elements: Point, Line, Plane, Volume. Point-ന് position മാത്രമേ ഉള്ളൂ. Line ഒരു moving point-ന്റെ path. Plane രണ്ട് dimension-ലേക്ക് extend ചെയ്യുന്നു. Volume മൂന്ന് dimension-ൽ mass നൽകുന്നു.

Point line plane and volume elements

Point
Position

Line

Direction + Length

Plane

2D Surface

Volume

3D Mass

Point line plane and volume elements. Identify parts, trace visual/spatial flow, then state one application and one design significance.

Architecture connection

Relate this concept to an actual building, spatial experience, material choice or cultural context. Clearly state the design intent, visual strategy and environmental response.

Self-check

Define it, identify or draw its main features, explain the design principle, compare one alternative and state an architectural application plus limitation.

1.3 Application of elements in architecture 3 h · Understand

The elements of design are not abstract theory — they are the tools with which architects shape space and meaning. A point becomes a column, a lamp, a sculpture or a threshold that orients the visitor. Lines become walls, cornices, mullions and railings that direct movement and frame views. Planes become floors, walls, ceilings and roofs that define enclosure and privacy. Volumes become rooms, halls, atria and courtyards that give the building its spatial identity. The architect selects, combines and manipulates these elements to achieve the desired emotional, functional and cultural effect. For example, the soaring verticals of a Gothic cathedral nave elevate the spirit, while the horizontal emphasis of a Frank Lloyd Wright prairie house anchors the building to the earth. Understanding these applications allows the student to analyse any building and identify the designer's visual strategy.

Point as focal element: column, sculpture, threshold, landmark

Line as directional element: walls, cornices, mullions, railings, paths

Plane as enclosing element: floors, walls, ceilings, roofs, screens

Volume as spatial element: rooms, halls, atria, courtyards, masses

Vertical emphasis creates aspiration (Gothic cathedral); horizontal creates grounding (Prairie house)

Elements combine to create rhythm, hierarchy and spatial narrative

Malayalam support: Elements architecture-ൽ apply ചെയ്യുന്നു: Point → column/sculpture, Line → wall/cornice, Plane → floor/ceiling, Volume → room/hall. Architect elements combine ചെയ്യുമ്പോൾ rhythm, hierarchy, spatial narrative ഉണ്ടാകുന്നു.

Architectural application of elements

Architectural application of elements. Identify parts, trace visual/spatial flow, then state one application and one design significance.

Architecture connection

Relate this concept to an actual building, spatial experience, material choice or cultural context. Clearly state the design intent, visual strategy and environmental response.

Self-check

Define it, identify or draw its main features, explain the design principle, compare one alternative and state an architectural application plus limitation.

1.4 Representation of elements: line types and visual impact 4 h · Apply

In architectural representation, different line types carry specific meaning and visual weight. Vertical lines communicate stability, strength and upward aspiration — they are the language of columns, towers and skyscrapers. Horizontal lines suggest rest, calm, stability and breadth — they define floor levels, cornices and the horizon line in landscape. Diagonal lines create tension, movement and dynamism — they are used in roofs, ramps and structural bracing. Zigzag lines produce energy, vibration and visual agitation. Curvilinear lines introduce softness, grace and organic character — they appear in arches, domes and flowing façades. Spiral lines suggest growth, evolution and infinite continuation. The student must learn to identify these line types in real buildings, sketch them accurately, and explain the visual and emotional impact each creates.

Vertical lines: stability, strength, aspiration
(columns, towers)

Horizontal lines: rest, calm, breadth (cornices,
floor lines, horizon)

Diagonal lines: tension, movement, dynamism
(roofs, ramps, bracing)

Zigzag lines: energy, vibration, agitation

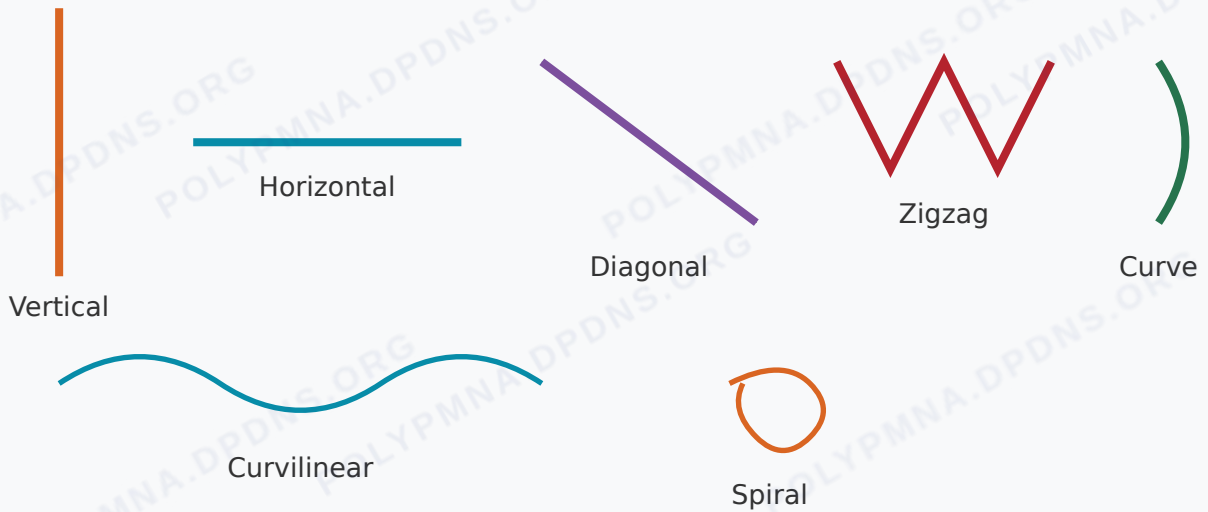
Curvilinear lines: softness, grace, organic flow
(arches, domes)

Spiral lines: growth, evolution, infinity

Each line type carries distinct visual and
emotional meaning

Malayalam support: Line types: Vertical → stability/aspiration, Horizontal → rest/calm, Diagonal → tension/movement, Zigzag → energy, Curvilinear → softness/grace, Spiral → growth. ഓരോ line type-നും distinct visual meaning ഉണ്ട്.

Line types and visual impact



Each line type carries distinct visual and emotional meaning

Line types and visual impact. Identify parts, trace visual/spatial flow, then state one application and one design significance.

Architecture connection

Relate this concept to an actual building, spatial experience, material choice or cultural context. Clearly state the design intent, visual strategy and environmental response.

Self-check

Define it, identify or draw its main features, explain the design principle, compare one alternative and state an architectural application plus limitation.

1.5 Representation: compositions of solids, voids, lighting and shading

4 h · Apply

Architectural representation extends beyond line types to the composition of solids and voids and the application of light and shadow. Solids represent the built mass — walls, columns, roofs and structural elements. Voids represent the negative space — openings, windows, courtyards, atria and the spaces between elements. The interplay of solid and void creates rhythm, proportion and visual interest. Lighting transforms this composition: directional light from a single source creates strong contrast with deep shadows that define form and depth; diffused light from multiple sources softens edges and reveals surface texture; back-lighting creates silhouettes that emphasise outline. Shading techniques include hatching (parallel lines at a consistent angle), cross-hatching (overlapping sets of parallel lines for darker tones), stippling (dots for gradual tonal transition), and smudging (blending for smooth gradients). Mastery of these techniques allows accurate representation of three-dimensional form on a two-dimensional sheet.

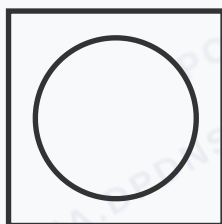
Solids: built mass — walls, columns, roofs, structural elements	Voids: negative space — openings, windows, courtyards, atria
Solid-void interplay creates rhythm, proportion and visual interest	Directional light: strong contrast, deep shadows, form definition
Diffused light: soft edges, texture revelation	Back-lighting: silhouette, outline emphasis
Shading techniques: hatching, cross-hatching, stippling, smudging	

Malayalam support: Solids → built mass, Voids → negative space. Solid-void interplay rhythm, proportion, visual interest രാസം. Lighting types: directional, diffused, back-lighting. Shading: hatching, cross-hatching, stippling, smudging.

Solid void composition and shading



Solid



Void



Shading

Solid = built mass · Void = negative space · Shading = depth

● Stippling

====
Hatching

Solid void composition and shading. Identify parts, trace visual/spatial flow, then state one application and one design significance.

Architecture connection

Relate this concept to an actual building, spatial experience, material choice or cultural context. Clearly state the design intent, visual strategy and environmental response.

Self-check

Define it, identify or draw its main features, explain the design principle, compare one alternative and state an architectural application plus limitation.

MODULE II · CO2

Colour and Texture

Theory 12 h

Practical 6 h

CO2

2.1 Colour theory 6 h · Understand

Colour is one of the most powerful tools in architectural design. It influences perception of space, mood, scale and temperature. The three qualities of colour are hue (the name of the colour, such as red or blue), value (the lightness or darkness of a colour), and intensity or chroma (the purity or saturation of a colour). Primary colours — red, yellow and blue — cannot be created by mixing other colours. Secondary colours — orange, green and violet — are created by mixing two primaries. Tertiary colours are created by mixing a primary with an adjacent secondary. The colour wheel organises these relationships and identifies complementary pairs (opposite on the wheel), analogous groups (adjacent), triadic sets (evenly spaced), and split-complementary schemes. Colour psychology associates specific emotional and cultural meanings with hues: red evokes energy and urgency, blue suggests calm and trust, green represents nature and balance, yellow conveys warmth and optimism, while neutral tones create sophistication and restraint. Warm colours (red, orange, yellow) appear to advance and make spaces feel intimate; cool colours (blue, green, violet) appear to recede and expand perceived space.

Three qualities: Hue (colour name), Value (lightness/darkness), Intensity/Chroma (purity/saturation)

Primary colours: red, yellow, blue — cannot be mixed from others

Secondary colours: orange, green, violet — from two primaries

Tertiary colours: from primary + adjacent secondary

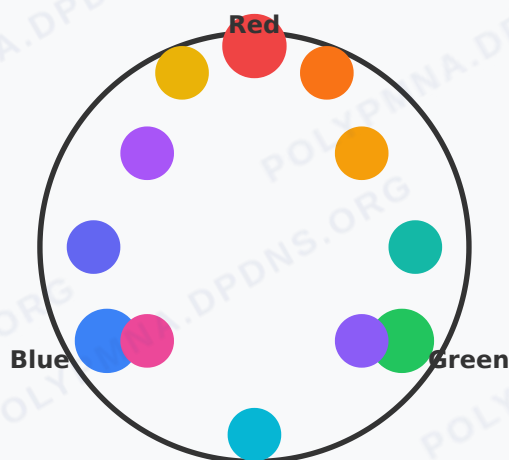
Colour wheel: complementary (opposite), analogous (adjacent), triadic (evenly spaced)

Colour psychology: red=energy, blue=calm, green=nature, yellow=warmth, neutrals=sophistication

Warm colours advance (intimate); cool colours recede (expand space)

Malayalam support: Colour-ന്റെ 3 qualities: Hue, Value, Intensity. Primary: red, yellow, blue. Secondary: orange, green, violet. Tertiary: primary + adjacent secondary. Colour wheel complementary, analogous, triadic schemes കാണിക്കുന്നു. Warm colours advance; cool colours recede.

Colour wheel and colour theory



Complementary: Red↔Green, Blue↔Orange, Yellow↔Violet

Primary: Red, Yellow, Blue · Secondary: Orange, Green, Violet

Tertiary: from primary + adjacent secondary

Colour wheel and colour theory. Identify parts, trace visual/spatial flow, then state one application and one design significance.

Architecture connection

Relate this concept to an actual building, spatial experience, material choice or cultural context. Clearly state the design intent, visual strategy and environmental response.

Self-check

Define it, identify or draw its main features, explain the design principle, compare one alternative and state an architectural application plus limitation.

2.2 Texture in architectural design 6 h · Understand

Texture is the perceived or actual surface quality of a material. It affects how light interacts with a surface, how the eye perceives scale and distance, and how the hand responds to touch. Actual (tactile) texture can be felt physically — rough brick, smooth marble, corrugated metal. Visual (optical) texture is implied through pattern, colour variation or photographic representation. Architectural textures serve multiple functions: they provide visual interest and break monotony, they communicate material honesty and construction method, they influence acoustic properties (rough surfaces absorb sound, smooth surfaces reflect), they affect thermal performance (dark rough surfaces absorb heat, light smooth surfaces reflect), and they establish cultural and contextual identity. Common architectural textures include rough (stone, exposed brick, bark), smooth (polished marble, glass, plaster), glossy (polished granite, lacquer), matte (chalk, unpolished concrete), patterned (tile mosaics, decorative grilles), and organic (wood grain, foliage shadows). The architect selects texture deliberately to enhance the design concept and respond to climate, function and cultural context.

Texture: perceived or actual surface quality of a material

Tactile (actual) texture: can be felt — rough brick, smooth marble

Visual (optical) texture: implied through pattern or colour variation

Functions: visual interest, material honesty, acoustic, thermal, cultural identity

Rough surfaces absorb sound; smooth surfaces reflect

Dark rough surfaces absorb heat; light smooth surfaces reflect

Common types: rough, smooth, glossy, matte, patterned, organic

Malayalam support: Texture surface-ന്റെ quality ആണ്. Tactile texture touch ചെയ്ത് feel ചെയ്യാം; visual texture pattern/colour വഴി implied. Texture functions: visual interest, material honesty, acoustic, thermal, cultural identity.

Texture types and architectural application



Rough (Stone)



Smooth (Marble)



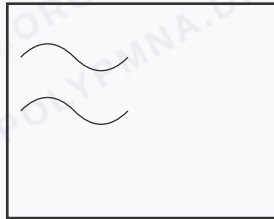
Glossy (Granite)



Matte



Patterned



Organic

Texture affects visual interest, acoustics, thermal performance and cultural identity

Rough absorbs sound/heat · Smooth reflects · Dark rough absorbs · Light smooth reflects

Texture types and architectural application. Identify parts, trace visual/spatial flow, then state one application and one design significance.

Architecture connection

Relate this concept to an actual building, spatial experience, material choice or cultural context. Clearly state the design intent, visual strategy and environmental response.

Self-check

Define it, identify or draw its main features, explain the design principle, compare one alternative and state an architectural application plus limitation.

2.3 Colour scheme and the colour wheel 4 h · Understand

A colour scheme is a deliberate selection of colours based on their relationship on the colour wheel. Understanding the colour wheel is essential for creating harmonious architectural palettes. The wheel organises hues by wavelength and reveals six primary relationship types. A monochromatic scheme uses one hue with variations in value and intensity — it creates unity and sophistication but risks monotony. An analogous scheme uses colours adjacent on the wheel — it produces harmony and ease but limited contrast. A complementary scheme uses colours opposite on the wheel — it creates maximum contrast and vibrancy but can be overwhelming if not balanced. A split-complementary scheme uses one hue plus the two hues adjacent to its complement — it offers strong contrast with less tension. A triadic scheme uses three hues evenly spaced — it provides balance and variety. A tetradic (double-complementary) scheme uses two complementary pairs — it is rich and complex but requires careful management of dominance.

Monochromatic: one hue, varied value/intensity — unity, sophistication

Analogous: adjacent hues — harmony, limited contrast

Complementary: opposite hues — maximum contrast, vibrancy

Split-complementary: one hue + two adjacent to complement — strong contrast, less tension

Triadic: three evenly spaced hues — balance and variety

Tetradic: two complementary pairs — rich, complex, needs dominance control

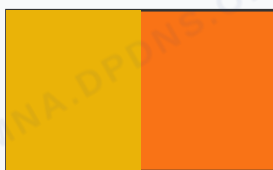
Warm/cool balance determines perceived temperature and spatial effect

Malayalam support: Colour scheme colour wheel-ന്റെ relationship അടിസ്ഥാനമാക്കി select ചെയ്യുന്നു. Monochromatic → unity, Analogous → harmony, Complementary → contrast, Triadic → balance, Tetradic → complex. Warm/cool balance spatial effect determine ചെയ്യുന്നു.

Colour wheel and scheme types



Monochromatic



Analogous



Complementary



Triadic

Warm Colours

Red, Orange, Yellow

Appear to ADVANCE

Create intimacy

Cool Colours

Blue, Green, Violet

Appear to RECEDE

Expand perceived space

Dominance (largest area) · Subordination (supporting) · Accent (focal contrast)

Colour wheel and scheme types. Identify parts, trace visual/spatial flow, then state one application and one design significance.

Architecture connection

Relate this concept to an actual building, spatial experience, material choice or cultural context. Clearly state the design intent, visual strategy and environmental response.

Self-check

Define it, identify or draw its main features, explain the design principle, compare one alternative and state an architectural application plus limitation.

2.4 Colour compositions: practical application 2 h · Apply

Colour composition is the practical arrangement of colours on a surface or within a space to achieve a specific visual and emotional effect. The student learns to create four standard composition types: complementary compositions use opposing colours for dramatic contrast and energy; analogous compositions use neighbouring hues for calm and unity; monochromatic compositions explore the full range of a single hue from light to dark for elegance and restraint; and triadic compositions distribute three equally spaced hues for balanced vibrancy. Each composition must demonstrate understanding of colour dominance (one colour occupies the largest area), subordination (a second colour supports), and accent (a small area of contrasting colour for focal emphasis). The compositions should be presented on standard drawing sheets with clear labels, a colour key, and a written explanation of the intended effect.

Complementary composition: opposing colours, dramatic contrast

Analogous composition: neighbouring hues, calm and unity

Monochromatic composition: single hue range, elegance and restraint

Triadic composition: three evenly spaced hues, balanced vibrancy

Colour dominance: one colour occupies largest area

Colour subordination: supporting colour

Colour accent: small contrasting area for focal emphasis

Each composition requires labels, colour key and written explanation

Malayalam support: Colour composition: surface-ൽ colours arrange ചെയ്യുമ്പോൾ intended effect ഉണ്ടാകുന്നു. Dominance (largest area), Subordination (supporting), Accent (focal contrast) എന്നിവ apply ചെയ്യണം.

Colour composition types

Figure reference — reproduce in examination drawings

Colour composition types. Identify parts, trace visual/spatial flow, then state one application and one design significance.

Architecture connection

Relate this concept to an actual building, spatial experience, material choice or cultural context. Clearly state the design intent, visual strategy and environmental response.

Self-check

Define it, identify or draw its main features, explain the design principle, compare one alternative and state an architectural application plus limitation.

2.5 Representation of texture 2 h · Apply

Texture representation in architectural drawing communicates the surface quality of materials without the actual material being present. The student prepares compositions using both natural materials (leaves, bark, stone fragments, fabric samples) and artificial materials (sandpaper grades, corrugated cardboard, wire mesh, textured paint samples). Each material is mounted on a drawing sheet, labelled with its name, source, tactile quality, visual character and typical architectural application. The student then creates drawn representations of these textures using appropriate techniques: cross-hatching for rough stone, parallel wavy lines for wood grain, stippling for sandy surfaces, smooth gradation for polished marble, and geometric patterns for manufactured tiles. This exercise develops the student's ability to observe, classify and represent surface qualities — a skill essential for architectural presentation and material specification.

Natural materials: leaves, bark, stone, fabric — tactile and visual study

Artificial materials: sandpaper, corrugated cardboard, wire mesh, textured paint

Mounting: each sample labelled with name, source, quality, character and application

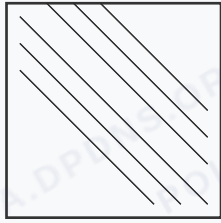
Drawn representation techniques: cross-hatching (stone), wavy lines (wood), stippling (sand), gradation (marble), geometric patterns (tiles)

Develops observation, classification and representation skills

Essential for architectural presentation and material specification

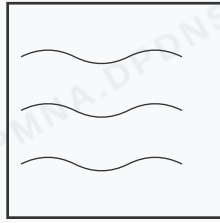
Malayalam support: Texture representation: natural, artificial materials mount ചെയ്യുക. Drawn techniques: cross-hatching → stone, wavy lines → wood, stippling → sand, gradation → marble. Observation, classification, representation skills develop ചെയ്യുന്നു.

Texture representation techniques



Cross-hatching

(Stone)



Wavy Lines

(Wood grain)



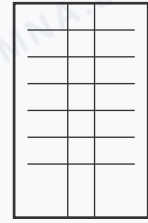
Stippling

(Sand/Sandstone)



Gradation

(Polished marble)



Geometric

(Tiles)

Each technique represents a different surface texture quality

Match technique to material: stone→cross-hatch, wood→wavy, sand→stipple, marble→gradation, tile→geometric

Texture representation techniques. Identify parts, trace visual/spatial flow, then state one application and one design significance.

Architecture connection

Relate this concept to an actual building, spatial experience, material choice or cultural context. Clearly state the design intent, visual strategy and environmental response.

Self-check

Define it, identify or draw its main features, explain the design principle, compare one alternative and state an architectural application plus limitation.

MODULE III · CO3

Principles of Design

Theory 12 h

Practical 10 h

CO3

3.1 Principles of design 9 h · Understand

The principles of design are the rules that govern how the elements of design are organised to create a successful composition. They are the grammar of visual language in architecture. Balance refers to the distribution of visual weight — it may be symmetrical (mirror-image about a central axis, conveying formality and stability), asymmetrical (different elements balanced by visual weight rather than identical shape, conveying dynamism), or radial (elements arranged around a central point, as in a dome or rotunda). Proportion and scale deal with the relationship between parts and the whole, and between the building and the human body. The golden ratio (1:1.618) and the Modulor system of Le Corbusier are mathematical foundations. Rhythm is the repetition or alternation of elements that creates visual movement — it may be regular (uniform spacing), flowing (curved progression), progressive (graduated change), alternating (two or more elements), or radiating (from a centre). Hierarchy establishes order of importance through size, position, colour contrast or complexity — the main entrance is always hierarchically dominant. Contrast creates visual interest through differences in size, shape, colour, texture or direction. Harmony is the pleasing combination of elements that share common characteristics. Emphasis creates a focal point that attracts the viewer's attention and anchors the composition.

Balance: symmetrical (formal), asymmetrical (dynamic), radial (centred)

Proportion and scale: relationship between parts, whole and human body; golden ratio 1:1.618

Rhythm: regular, flowing, progressive, alternating, radiating

Hierarchy: order of importance through size, position, colour, complexity

Contrast: visual interest through differences in size, shape, colour, texture, direction

Harmony: pleasing combination of elements sharing common characteristics

Emphasis: focal point that anchors the composition

Malayalam support: Design principles: Balance (symmetrical/asymmetrical/radial), Proportion & Scale (golden ratio 1:1.618), Rhythm (regular/flowing/progressive/alternating/radiating), Hierarchy (importance order), Contrast (differences), Harmony (pleasing combination), Emphasis (focal point).

Principles of design

Principles of design. Identify parts, trace visual/spatial flow, then state one application and one design significance.

Architecture connection

Relate this concept to an actual building, spatial experience, material choice or cultural context. Clearly state the design intent, visual strategy and environmental response.

Self-check

Define it, identify or draw its main features, explain the design principle, compare one alternative and state an architectural application plus limitation.

3.2 Illustration of principles in architecture 3 h · Understand

The principles of design are not abstract — they are visible in every great building. Balance is evident in the symmetrical façades of the Taj Mahal and Rashtrapathi Bhavan. Proportion and scale govern the relationship between the massive dome and the human-scaled courtyard in Sangath. Rhythm is expressed through the colonnade of the Greek Parthenon and the repetitive bays of a Gothic cathedral. Hierarchy is demonstrated by the dominant entrance portal of Falling Water and the subsidiary guest wings. Contrast appears in Le Corbusier's Villa Savoye through the white rendered surface against the dark pilotis and green landscape. Harmony unifies the warm sandstone palette of Sangath. Emphasis directs the eye to the central rotunda of Rashtrapathi Bhavan. By identifying these principles in real buildings, the student develops the critical eye needed to analyse and create architecture.

Taj Mahal: symmetrical balance, golden proportion, hierarchical emphasis

Sangath (Doshi): human scale, rhythmic vaults, harmonious materials

Rashtrapathi Bhavan (Lutyens): axial hierarchy, monumental scale, classical rhythm

Falling Water (Wright): organic balance, horizontal emphasis, contrast with nature

Villa Savoye (Le Corbusier): proportional pilotis, contrast of white and dark, emphasis on horizontality

Farnsworth House (Mies): minimal hierarchy, glass-wall transparency, rhythmic column grid

Critical eye: analyse real buildings to identify and understand principles

Malayalam support: Great buildings-ൽ design principles visible ആണ്. Taj Mahal → symmetrical balance; Sangath → human scale, rhythmic vaults; Rashtrapathi Bhavan → axial hierarchy; Falling Water → organic balance; Villa Savoye → proportional pilotis; Farnsworth → minimal hierarchy. Real buildings analyse ചെയ്ത് critical eye develop ചെയ്യുക.

Design principles in famous buildings

Design principles in famous buildings. Identify parts, trace visual/spatial flow, then state one application and one design significance.

Architecture connection

Relate this concept to an actual building, spatial experience, material choice or cultural context. Clearly state the design intent, visual strategy and environmental response.

Self-check

Define it, identify or draw its main features, explain the design principle, compare one alternative and state an architectural application plus limitation.

3.3 Representation of principles through design compositions 6 h · Apply

The student creates comprehensive design compositions that demonstrate the application of each principle through controlled drawing exercises. A balance composition shows symmetrical, asymmetrical and radial arrangements using geometric shapes, with annotations explaining how visual weight is distributed. A proportion and scale composition demonstrates the golden ratio using a spiral construction, applies the Modulator grid to a human figure within a room, and shows how changing the scale of a single element alters the entire composition. A rhythm composition arranges elements in regular, flowing, progressive, alternating and radiating sequences. A hierarchy composition uses size, position and colour to establish a clear order of importance among five or more elements. A contrast composition juxtaposes opposing qualities. A harmony composition unifies diverse elements through shared characteristics. An emphasis composition creates a single dominant focal point within a field of supporting elements. Each composition must be labelled, annotated and accompanied by a brief written explanation.

Balance composition: symmetrical, asymmetrical, radial with annotations

Proportion composition: golden ratio spiral, Modulator grid, scale comparison

Rhythm composition: regular, flowing, progressive, alternating, radiating sequences

Hierarchy composition: size, position, colour to establish importance order

Contrast composition: opposing qualities juxtaposed

Harmony composition: diverse elements unified by shared characteristics

Emphasis composition: single dominant focal point within supporting field

All compositions: labelled, annotated, written explanation

Malayalam support: Design compositions ഉണ്ടാക്കുമ്പോൾ ഓരോ principle demonstrate ചെയ്യണം. Balance → symmetrical/asymmetrical/radial. Proportion → golden ratio spiral, Modulator. Rhythm → 5 types. Hierarchy → size, position, colour. Contrast, Harmony, Emphasis എന്നിവയും compositions വഴി show ചെയ്യുക.

Design composition exercises

Design composition exercises. Identify parts, trace visual/spatial flow, then state one application and one design significance.

Architecture connection

Relate this concept to an actual building, spatial experience, material choice or cultural context. Clearly state the design intent, visual strategy and environmental response.

Self-check

Define it, identify or draw its main features, explain the design principle, compare one alternative and state an architectural application plus limitation.

3.4 Freehand sketching of outdoor spaces 4 h · Apply

Freehand sketching is the most direct means of architectural observation and communication. Unlike instrument drawing, it captures the living quality of a space — the play of light, the rhythm of people, the texture of surfaces and the atmosphere of a place. The student selects a nearby outdoor space — a courtyard, street scene, garden, market or public building — and produces a series of sketches. The first sketch is a quick gesture drawing (30 seconds) that captures the overall impression and dominant lines. The second is a study sketch (3-5 minutes) that develops the major shapes and spatial relationships. The third is a finished sketch (15-30 minutes) that applies appropriate lighting and shading concepts to represent depth, form and spatial quality. The student must identify and label the architectural elements visible in the sketch, note the direction and quality of light, and write a brief commentary on the design principles observed. This practice develops the hand-eye coordination, observational accuracy and analytical thinking essential for architectural design.

Gesture drawing (30s): overall impression, dominant lines, energy

Study sketch (3-5 min): major shapes, spatial relationships, composition

Finished sketch (15-30 min): lighting, shading, depth, form, spatial quality

Identify and label architectural elements in the sketch

Note direction and quality of light (directional, diffused, ambient)

Write commentary on design principles observed

Develops hand-eye coordination, observational accuracy, analytical thinking

Essential skill for architectural design and communication

Malayalam support: Freehand sketching architectural observation-ന്റെ most direct method. Gesture drawing → overall impression. Study sketch → shapes, relationships. Finished sketch → lighting, shading, depth. Elements identify, label ചെയ്യുക. Design principles commentary എഴുതുക. Hand-eye coordination, observational accuracy develop ചെയ്യുന്നു.

Freehand sketching process

Freehand sketching process. Identify parts, trace visual/spatial flow, then state one application and one design significance.

Architecture connection

Relate this concept to an actual building, spatial experience, material choice or cultural context. Clearly state the design intent, visual strategy and environmental response.

Self-check

Define it, identify or draw its main features, explain the design principle, compare one alternative and state an architectural application plus limitation.

MODULE IV · CO4

Famous Architectural Buildings

Theory 18 h

Practical 12 h

CO4

4.1 Introduction of famous Indian buildings 6 h · Understand

Indian architecture spans ancient temple traditions, Mughal monuments, colonial structures and modernist masterpieces. This module examines three landmark projects that define different eras of Indian architectural thought. Kovalam Beach Resort (Charles Correa) is a masterwork of tropical modernism that responds to Kerala's coastal climate through a series of low-rise pavilions arranged along the beach, with sloping tiled roofs, deep verandahs and natural ventilation. The design celebrates local materials (terracotta, wood, laterite stone) and traditional Kerala spatial organisation. Sangath (B.V. Doshi), Doshi's own studio in Ahmedabad, is a deeply personal work that integrates Indian vernacular with modernist abstraction — vaulted brick shells create cave-like interiors cooled by natural ventilation, sunken gardens and water channels respond to the harsh Gujarat climate, and the irregular plan encourages creative movement and spontaneous collaboration. Rashtrapathi Bhavan (Edwin Lutyens), the presidential residence in New Delhi, is a monumental synthesis of classical European architecture and Indian decorative tradition — the massive dome (third-largest in the world) crowns a rectangular mass adorned with Indian lotus motifs, Mughal chattris and classical columns, expressing imperial authority while acknowledging the host culture.

Kovalam Beach Resort (Correa): tropical modernism, coastal climate response, local materials, traditional Kerala organisation

Sangath (Doshi): Indian vernacular + modernist abstraction, vaulted shells, natural ventilation, sunken gardens, creative spatial organisation

Rashtrapathi Bhavan (Lutyens): classical European + Indian decorative, massive dome, lotus motifs, Mughal chattris, imperial authority

Each building reflects its architect's philosophy, climate response and cultural context

Study: design concept, spatial organisation, material selection, climate response, significance

Malayalam support: Indian buildings: Kovalam → tropical modernism, local materials. Sangath → vernacular + modernist, vaulted shells, natural cooling. Rashtrapathi Bhavan → classical + Indian, massive dome, lotus motifs. ഈ മൂന്ന് കെട്ടിടങ്ങളുടെയും ഓരോ കെട്ടിടത്തിന്റെയും ഏകദേശം, കാലാവസ്ഥാ പ്രതികരണം, സാമൂഹിക പശ്ചാത്തലം പ്രതിഫലിപ്പിക്കുന്നു.

Famous Indian architectural buildings

Famous Indian architectural buildings. Identify parts, trace visual/spatial flow, then state one application and one design significance.

Architecture connection

Relate this concept to an actual building, spatial experience, material choice or cultural context. Clearly state the design intent, visual strategy and environmental response.

Self-check

Define it, identify or draw its main features, explain the design principle, compare one alternative and state an architectural application plus limitation.

4.2 Design features and significance of Indian buildings 2 h · Understand

The architectural significance of these Indian buildings extends beyond aesthetics to encompass philosophy, sustainability and cultural identity. Charles Correa's philosophy centres on the belief that architecture must grow from the land and the people — his buildings respond to climate, culture and craft. Kovalam demonstrates how modern architecture can honour tradition without imitating it. B.V. Doshi's Sangath embodies his lifelong commitment to sustainable, community-centred architecture — the building is a living manifesto that proves comfort can be achieved without mechanical systems, using ancient Indian passive cooling strategies reinterpreted through modern form. The architecture of Rashtrapathi Bhavan represents a pivotal moment in colonial history: Lutyens deliberately incorporated Indian motifs into a European classical framework, creating a hybrid that acknowledged Indian culture while asserting imperial presence. This synthesis — controversial then, historically significant now — demonstrates how architecture can negotiate between cultures, between tradition and modernity, between the universal and the local.

Correa: architecture grows from land and people; tradition honoured without imitation

Doshi: sustainable, community-centred; passive cooling without mechanical systems; ancient strategies reinterpreted

Lutyens: colonial Indian synthesis; Indian motifs within European classical framework

Cultural negotiation: between tradition and modernity, universal and local

Historical significance: architecture as cultural document and political statement

Material honesty: local materials, local craft, local climate response

Malayalam support: Design features: Correa → land-ൽ നിന്ന് grow ചെയ്യണം; tradition honour ചെയ്യുക but imitate അല്ല. Doshi → sustainable, passive cooling, ancient strategies modern form-ൽ. Lutyens → colonial synthesis, Indian motifs European classical-ൽ. Architecture culture negotiate ചെയ്യുന്നു.

Indian buildings design features

Indian buildings design features. Identify parts, trace visual/spatial flow, then state one application and one design significance.

Architecture connection

Relate this concept to an actual building, spatial experience, material choice or cultural context. Clearly state the design intent, visual strategy and environmental response.

Self-check

Define it, identify or draw its main features, explain the design principle, compare one alternative and state an architectural application plus limitation.

4.3 Freehand sketching of Indian buildings 6 h · Apply

The student produces freehand sketches of the three Indian buildings studied, using photographs and reference images as guides. Each sketch must capture the distinctive architectural features identified during the lecture and tutorial sessions. For Kovalam Beach Resort, the sketch should show the low-rise pavilion arrangement along the beach, the sloping tiled roofs, the deep verandahs and the relationship between built form and landscape. For Sangath, the sketch should capture the irregular plan, the vaulted brick shells, the sunken garden spaces and the interplay of solid and void. For Rashtrapathi Bhavan, the sketch should show the axial symmetry, the massive dome, the colonnade and the integration of Indian decorative elements. Each sketch must include a caption identifying the building, architect, location and date, followed by a short paragraph explaining the design concept and the student's interpretation of the architectural significance. The sketches demonstrate the student's ability to observe, analyse and represent architectural form through manual drawing.

Kovalam sketch: pavilion arrangement, sloping roofs, verandahs, beach relationship

Sangath sketch: irregular plan, vaulted shells, sunken gardens, solid-void interplay

Rashtrapathi Bhavan sketch: axial symmetry, dome, colonnade, Indian decorative elements

Each sketch: caption with building name, architect, location, date

Written paragraph: design concept and student interpretation

Demonstrates observation, analysis and manual representation skills

Use photographs and references as guides, not for tracing

Malayalam support: Indian buildings sketch ചെയ്യുക: Kovalam → pavilion arrangement, roofs, verandahs, beach. Sangath → irregular plan, vaulted shells, sunken gardens. Rashtrapathi Bhavan → symmetry, dome, colonnade. Caption, written explanation ഉൾപ്പെടുത്തുക. Manual drawing skills demonstrate ചെയ്യുക.

Indian buildings sketching

Indian buildings sketching. Identify parts, trace visual/spatial flow, then state one application and one design significance.

Architecture connection

Relate this concept to an actual building, spatial experience, material choice or cultural context. Clearly state the design intent, visual strategy and environmental response.

Self-check

Define it, identify or draw its main features, explain the design principle, compare one alternative and state an architectural application plus limitation.

4.4 Introduction of famous International buildings 6 h · Understand

International architecture of the twentieth century revolutionised how buildings relate to nature, structure and human life. Falling Water (Frank Lloyd Wright, 1935, Pennsylvania, USA) is built over a waterfall in the Allegheny Mountains and exemplifies organic architecture — the building grows from the rock, cantilevered concrete terraces extend over the stream, and the horizontal lines echo the natural rock strata. The interior is centred on the natural rock hearth, blurring the boundary between inside and outside. Villa Savoye (Le Corbusier, 1929, Poissy, France) is the definitive manifesto of the International Style and the Five Points of Architecture: pilotis (reinforced concrete columns) lift the building above the ground, the free façade allows windows of any shape, the open plan eliminates load-bearing walls, the ribbon window provides uniform light, and the roof garden restores the land taken by the building footprint. The pure white rendered surface and geometric purity express a machine-age aesthetic. Farnsworth House (Mies Van der Rohe, 1951, Illinois, USA) is the ultimate expression of architectural minimalism — a glass box elevated on eight white steel columns, floating above a meadow beside the Fox River. The transparent walls dissolve the boundary between interior and landscape, and the single-room plan contains only a central core of kitchen, bathroom and fireplace.

Falling Water (Wright, 1935): organic architecture, built over waterfall, cantilevered terraces, rock hearth, inside-outside blur

Villa Savoye (Le Corbusier, 1929): Five Points — pilotis, free façade, open plan, ribbon window, roof garden; International Style manifesto

Farnsworth House (Mies, 1951): architectural minimalism, glass box on steel columns, transparent walls, single-room plan, central core

Each building represents a paradigm shift in architectural thought

Study: design concept, spatial organisation, material, climate response, significance

Malayalam support: International buildings: Falling Water → organic architecture, waterfall-ൽ, cantilevered terraces. Villa Savoye → Five Points, International Style, machine-age aesthetic. Farnsworth House → minimalism, glass box, transparent walls. ഈ രണ്ടു building paradigm shift represent ചെയ്യുന്നു.

Famous International architectural buildings

Famous International architectural buildings. Identify parts, trace visual/spatial flow, then state one application and one design significance.

Architecture connection

Relate this concept to an actual building, spatial experience, material choice or cultural context. Clearly state the design intent, visual strategy and environmental response.

Self-check

Define it, identify or draw its main features, explain the design principle, compare one alternative and state an architectural application plus limitation.

4.5 Design features and significance of International buildings 2 h · Understand

These three international buildings represent three distinct architectural philosophies that have shaped the global discourse. Wright's organic architecture argues that a building should belong to its site as naturally as a plant belongs to the soil — every line, material and proportion responds to the specific landscape, climate and culture. Villa Savoye represents Le Corbusier's conviction that architecture must embrace the machine age: industrial materials (reinforced concrete, steel, glass), rational planning based on the Modulor, and a universal aesthetic that transcends local tradition. The building is a 'machine for living in' — functional, efficient and beautiful through geometric purity. Farnsworth House represents Mies Van der Rohe's pursuit of 'less is more' — the elimination of all non-essential elements to reveal the essential relationship between structure, space and nature. The building is simultaneously a dwelling, a sculpture and a lens through which the landscape is experienced. Together, these buildings demonstrate that architecture is not merely construction — it is philosophy made visible in built form.

Wright: building belongs to site like plant to soil; site-specific response

Le Corbusier: machine age, industrial materials, Modulor, universal aesthetic, 'machine for living'

Mies Van der Rohe: 'less is more', elimination of non-essential, structure-space-nature relationship

Philosophy: architecture as visible philosophy in built form

Material innovation: reinforced concrete, steel, glass as architectural media

Universal vs. site-specific: two poles of modernist architectural thought

Malayalam support: International buildings philosophies: Wright → site-specific organic, Le Corbusier → machine age universal, Mies → 'less is more'. Architecture philosophy made visible ആണ്. Material innovation: concrete, steel, glass. Universal vs site-specific two poles.

International buildings design features

International buildings design features. Identify parts, trace visual/spatial flow, then state one application and one design significance.

Architecture connection

Relate this concept to an actual building, spatial experience, material choice or cultural context. Clearly state the design intent, visual strategy and environmental response.

Self-check

Define it, identify or draw its main features, explain the design principle, compare one alternative and state an architectural application plus limitation.

4.6 Freehand sketching of International buildings 6 h · Apply

The student produces freehand sketches of the three International buildings, demonstrating understanding of their distinctive architectural features. For Falling Water, the sketch should capture the dramatic cantilevered terraces extending over the waterfall, the horizontal emphasis, the integration of rock and water into the composition, and the relationship between the built form and the forested landscape. For Villa Savoye, the sketch should show the pilotis lifting the white box above the green field, the ribbon windows wrapping the upper level, the ramp leading to the roof garden, and the clear geometric composition. For Farnsworth House, the sketch should convey the ethereal quality of the glass box floating on steel columns, the transparency of the walls revealing the meadow beyond, the single central core, and the building's relationship to the river and trees. Each sketch must include a caption with building name, architect, location and year, followed by a paragraph explaining the design philosophy and the student's interpretation of its significance.

Falling Water sketch: cantilevered terraces, waterfall, horizontal lines, rock-water integration

Villa Savoye sketch: pilotis, white box, ribbon windows, ramp, roof garden, geometric composition

Farnsworth House sketch: glass box, steel columns, transparency, meadow, central core

Each sketch: caption (name, architect, location, year) + written paragraph

Design philosophy explanation and student interpretation

Demonstrates observational skill, analytical understanding and manual drawing ability

Use reference images for study, not for tracing

Malayalam support: International buildings sketch ചെയ്യുക: Falling Water → cantilevered terraces, waterfall, horizontal. Villa Savoye → pilotis, white box, ribbon windows, ramp. Farnsworth → glass box, steel columns, transparency. Caption + paragraph ഉൾപ്പെടുത്തുക. Observation, analysis, drawing skills demonstrate ചെയ്യുക.

International buildings sketching

International buildings sketching. Identify parts, trace visual/spatial flow, then state one application and one design significance.

Architecture connection

Relate this concept to an actual building, spatial experience, material choice or cultural context. Clearly state the design intent, visual strategy and environmental response.

Self-check

Define it, identify or draw its main features, explain the design principle, compare one alternative and state an architectural application plus limitation.

PRACTICUM

Practical identification and record centre

The official practical syllabus is expanded into safe observation records. These are not operating instructions for unsupervised work.

Faculty must isolate the equipment and demonstrate safe access. Students may identify, trace, sketch and record only within the authorised task.

Completion planner

- ☐ M1 · Activity 1: Draw sheet layout and practise line weights
- ☐ M1 · Activity 2: Demonstrate line types and their visual impact
- ☐ M1 · Activity 3: Draw solids and voids composition with shading
- ☐ M2 · Activity 4: Prepare colour wheel and identify schemes
- ☐ M2 · Activity 5: Create texture compositions with natural and artificial materials
- ☐ M3 · Activity 6: Develop design compositions demonstrating principles
- ☐ M3 · Activity 7: Freehand sketch of an outdoor space
- ☐ M4 · Activity 8: Freehand sketch of famous Indian buildings
- ☐ M4 · Activity 9: Freehand sketch of famous International buildings

0 of 9 activities marked complete.

Activity 1: Draw sheet layout and practise line weights M1 · 4 h

Malayalam practical guide: Sheet-ൽ border, title block, drawing area correct ആയി lay out ചെയ്യുക. Line weights thick, medium, thin demonstrate ചെയ്യുക.

Aim

Prepare a standard A3 drawing sheet with correct border, title block, and demonstrate thick, medium and thin line weights.

Items / materials

A3 drawing sheet · Drawing board and T-square · Pencils (2H, HB, 2B) · Set squares · Eraser and sharpener

Mapped outcome

CO1 · CO1

Date / studio session

Faculty-supervised observation sequence

1. Secure sheet to board with tape
2. Draw border at 10mm from each edge
3. Create title block at bottom-right (60mm × 20mm)
4. Draw lines in three weight categories: thick (0.7mm), medium (0.5mm), thin (0.3mm)
5. Label each line type with its architectural meaning

6. Submit for faculty review

Record table

Element / feature	Identification	Design principle	Application	Visual impact

Observation focus: Sheet layout accuracy, line weight consistency, title block completeness, neatness.

Expected result: A correctly laid-out architectural drawing sheet with demonstrated line weight hierarchy.

Safety: Use pencil sharpener safely; keep work area clean.

Viva with answer cues

1. What feature identifies this element? *State visual characteristics and architectural role.*
2. What design principle does this demonstrate? *Name and explain the principle.*
3. What is the practical application in architecture? *Describe a real building example.*

Faculty signature: _____

Activity 2: Demonstrate line types and their visual impact M1 · 4 h

Malayalam practical guide: ഓരോ line type separate zone-ൽ draw ചെയ്യുക. Visual impact annotation-ൽ explain ചെയ്യുക. Architecture examples reference ആയി use ചെയ്യുക.

Aim

Create a composition showing vertical, horizontal, diagonal, zigzag, curvilinear and spiral lines with annotations explaining visual impact.

Items / materials

Drawing sheet · Pencils (2H, HB) · Eraser · Ruler for annotation lines

Mapped outcome

CO1 · CO1

Date / studio session

Faculty-supervised observation sequence

1. Divide sheet into six zones
2. Draw each line type in its zone
3. Add annotation arrows and text
4. Explain the visual and emotional impact of each line type
5. Compare examples from real architecture
6. Submit composition

Record table

Element / feature	Identification	Design principle	Application	Visual impact

Observation focus: Line type accuracy, annotation clarity, visual impact explanation, composition quality.

Expected result: A complete line-type composition with clear annotations demonstrating understanding of visual impact.

Safety: Maintain clean drawing surface; avoid smudging.

Viva with answer cues

1. What feature identifies this element? *State visual characteristics and architectural role.*
2. What design principle does this demonstrate? *Name and explain the principle.*
3. What is the practical application in architecture? *Describe a real building example.*

Faculty signature: _____

Activity 3: Draw solids and voids composition with shading
M1 · 4 h

Malayalam practical guide: Solid-void composition design ചെയ്യുക. Shading techniques: hatching, cross-hatching, stippling, smudging apply ചെയ്യുക. Light direction arrows add ചെയ്യുക.

Aim

Create a composition demonstrating solid-void relationships and apply hatching, cross-hatching, stippling and smudging techniques.

Items / materials

Drawing sheet · Pencils (HB, 2B, 4B) · Eraser · Tortillon or tissue for blending

Mapped outcome

CO1 · CO1

Date / studio session

Faculty-supervised observation sequence

1. Design a composition with clear solid and void areas
2. Draw solid masses first
3. Define void openings
4. Apply hatching for shadow on solids
5. Use cross-hatching for deeper shadows
6. Apply stippling for texture transition
7. Use smudging for smooth gradients
8. Add direction-of-light arrows
9. Submit with technique labels

Record table

Element / feature	Identification	Design principle	Application	Visual impact

Observation focus: Solid-void clarity, shading technique correctness, light direction consistency, composition balance.

Expected result: A well-composed solid-void drawing with correctly applied shading techniques.

Safety: Keep blending tools clean; avoid excessive pressure on paper.

Viva with answer cues

1. What feature identifies this element? *State visual characteristics and architectural role.*
2. What design principle does this demonstrate? *Name and explain the principle.*

3. What is the practical application in architecture? *Describe a real building example.*

Faculty signature: _____

Activity 4: Prepare colour wheel and identify schemes M2 · 2 h

Malayalam practical guide: 12-segment colour wheel ഉണ്ടാക്കുക. Primary, secondary, tertiary colours mix ചെയ്ത് place ചെയ്യുക. Complementary, analogous, triadic schemes identify ചെയ്ത് label ചെയ്യുക.

Aim

Create an accurate colour wheel showing primary, secondary and tertiary colours, then identify complementary, analogous and triadic schemes.

Items / materials

Drawing sheet · Watercolour or poster colour set · Colour wheel template or compass · Brushes · Pencil and eraser

Mapped outcome

CO2 · CO2

Date / studio session

Faculty-supervised observation sequence

1. Draw a circle divided into 12 segments
2. Place three primary colours at equal intervals
3. Mix and place secondary colours between primaries
4. Mix and place tertiary colours
5. Identify complementary pairs (opposite)
6. Identify analogous groups (adjacent three)
7. Identify triadic sets (every fourth)
8. Label each colour and scheme type
9. Submit with colour key

Record table

Element / feature	Identification	Design principle	Application	Visual impact

Observation focus: Colour accuracy, wheel geometry, scheme identification, labelling completeness.

Expected result: An accurate 12-segment colour wheel with correctly identified colour schemes.

Safety: Keep water containers stable; clean brushes after use.

Viva with answer cues

1. What feature identifies this element? *State visual characteristics and architectural role.*
2. What design principle does this demonstrate? *Name and explain the principle.*
3. What is the practical application in architecture? *Describe a real building example.*

Faculty signature: _____

Activity 5: Create texture compositions with natural and artificial materials

M2 · 2 h

Malayalam practical guide: Natural, artificial materials collect ചെയ്യുക. Mount ചെയ്യുക. Label: name, source, quality, application. Drawn representation techniques apply ചെയ്യുക.

Aim

Prepare compositions using natural and artificial texture materials, mounted and labelled with drawn representations.

Items / materials

Natural materials: leaves, bark, stone, fabric · Artificial materials: sandpaper, corrugated cardboard, wire mesh · Drawing sheet · Adhesive · Pencils · Labels

Mapped outcome

CO2 · CO2

Date / studio session

Faculty-supervised observation sequence

1. Collect and select materials
2. Mount each material on drawing sheet
3. Label with name, source, tactile quality and application
4. Create drawn representation of each texture using appropriate technique
5. Cross-hatching for rough, wavy lines for wood, stippling for granular
6. Submit complete texture portfolio

Record table

Element / feature	Identification	Design principle	Application	Visual impact

Observation focus: Material selection variety, mounting quality, labelling accuracy, drawn technique correctness.

Expected result: A complete texture portfolio with mounted samples and accurate drawn representations.

Safety: Handle sharp materials carefully; use adhesive in ventilated area.

Viva with answer cues

1. What feature identifies this element? *State visual characteristics and architectural role.*
2. What design principle does this demonstrate? *Name and explain the principle.*
3. What is the practical application in architecture? *Describe a real building example.*

Faculty signature: _____

Activity 6: Develop design compositions demonstrating principles M3 · 6 h

Malayalam practical guide: 3 sheets ഉണ്ടാക്കുക: Sheet 1 → Balance (symmetrical, asymmetrical, radial). Sheet 2 → Proportion, Rhythm. Sheet 3 → Hierarchy, Contrast, Harmony, Emphasis. Annotate ചെയ്യുക.

Aim

Create comprehensive design compositions demonstrating balance, proportion, rhythm, hierarchy, contrast, harmony and emphasis.

Items / materials

Drawing sheets (A3, 3 sheets) · Pencils · Erasers · Set squares · Rulers

Mapped outcome

CO3 · CO3

Date / studio session

Faculty-supervised observation sequence

1. Sheet 1: Balance composition — symmetrical, asymmetrical, radial arrangements
2. Sheet 2: Proportion and rhythm compositions — golden ratio, Modulor, five rhythm types
3. Sheet 3: Hierarchy, contrast, harmony, emphasis compositions
4. Annotate each composition with principle name and explanation
5. Check against principle definitions
6. Submit all three sheets

Record table

Element / feature	Identification	Design principle	Application	Visual impact

Observation focus: Principle demonstration accuracy, composition quality, annotation clarity, overall presentation.

Expected result: Three design composition sheets demonstrating all seven principles of design.

Safety: Keep drawing instruments sharp and safe.

Viva with answer cues

1. What feature identifies this element? *State visual characteristics and architectural role.*
2. What design principle does this demonstrate? *Name and explain the principle.*
3. What is the practical application in architecture? *Describe a real building example.*

Activity 7: Freehand sketch of an outdoor space M3 · 4 h

Malayalam practical guide: Outdoor space visit ചെയ്യുക. Observe → Gesture (30s) → Study (3-5 min) → Finished (15-30 min). Elements label ചെയ്യുക. Design principles commentary എഴുതുക.

Aim

Sketch a nearby outdoor space using gesture, study and finished techniques with appropriate lighting and shading.

Items / materials

Sketchbook or loose sheets · Pencils (2H, HB, 2B, 4B) · Eraser · Observation site (courtyard, street, garden or public space)

Mapped outcome

CO3 · CO3

Date / studio session

Faculty-supervised observation sequence

1. Visit selected outdoor space
2. Observe for 5 minutes: dominant lines, light direction, spatial quality
3. Gesture drawing (30 seconds): overall impression
4. Study sketch (3-5 minutes): major shapes and relationships
5. Finished sketch (15-30 minutes): lighting, shading, depth, form
6. Label architectural elements visible
7. Note light direction and quality
8. Write brief commentary on design principles observed
9. Submit sketch with commentary

Record table

Element / feature	Identification	Design principle	Application	Visual impact

Observation focus: Gesture energy, study accuracy, finished detail, element labelling, principle commentary.

Expected result: A series of sketches (gesture, study, finished) of an outdoor space with labelled elements and design commentary.

Safety: Be aware of surroundings when sketching outdoors; do not block pathways.

Viva with answer cues

1. What feature identifies this element? *State visual characteristics and architectural role.*
2. What design principle does this demonstrate? *Name and explain the principle.*
3. What is the practical application in architecture? *Describe a real building example.*

Faculty signature: _____

Activity 8: Freehand sketch of famous Indian buildings M4 · 6 h

Malayalam practical guide: Indian buildings 3 sketch ചെയ്യുക: Kovalam, Sangath, Rashtrapathi Bhavan. Caption: name, architect, location, date. Design commentary paragraph എഴുതുക.

Aim

Produce freehand sketches of Kovalam Beach Resort, Sangath and Rashtrapathi Bhavan with captions and design commentary.

Items / materials

Sketchbook or loose sheets · Pencils · Erasers · Reference photographs or images · Ruler for caption lines

Mapped outcome

CO4 · CO4

Date / studio session

Faculty-supervised observation sequence

1. Study reference images of each building
2. Kovalam: sketch pavilion arrangement, sloping roofs, beach relationship
3. Sangath: sketch irregular plan, vaulted shells, sunken gardens
4. Rashtrapathi Bhavan: sketch axial symmetry, dome, colonnade, Indian elements
5. Add caption: name, architect, location, date

6. Write paragraph explaining design concept and significance
7. Submit three sketches with commentary

Record table

Element / feature	Identification	Design principle	Application	Visual impact

Observation focus: Feature accuracy, composition quality, caption completeness, design commentary depth.

Expected result: Three freehand sketches of Indian buildings with captions and design commentary.

Safety: Use reference images for study only; do not trace.

Viva with answer cues

1. What feature identifies this element? *State visual characteristics and architectural role.*
2. What design principle does this demonstrate? *Name and explain the principle.*
3. What is the practical application in architecture? *Describe a real building example.*

Faculty signature: _____

Activity 9: Freehand sketch of famous International buildings M4 · 6 h

Malayalam practical guide: International buildings 3 sketch ചെങ്കുട്ടി: Falling Water, Villa Savoye, Farnsworth House. Caption: name, architect, location, year. Design philosophy commentary എഴുതുക.

Aim

Produce freehand sketches of Falling Water, Villa Savoye and Farnsworth House with captions and design commentary.

Items / materials

Sketchbook or loose sheets · Pencils · Erasers · Reference photographs or images · Ruler for caption lines

Mapped outcome

Date / studio session

Faculty-supervised observation sequence

1. Study reference images of each building
2. Falling Water: sketch cantilevered terraces, waterfall, horizontal lines
3. Villa Savoye: sketch pilotis, white box, ribbon windows, ramp
4. Farnsworth House: sketch glass box, steel columns, transparency, landscape
5. Add caption: name, architect, location, year
6. Write paragraph explaining design philosophy and significance
7. Submit three sketches with commentary

Record table

Element / feature	Identification	Design principle	Application	Visual impact

Observation focus: Feature accuracy, architectural understanding, caption completeness, philosophy explanation.

Expected result: Three freehand sketches of International buildings with captions and design commentary.

Safety: Use reference images for study only; do not trace.

Viva with answer cues

1. What feature identifies this element? *State visual characteristics and architectural role.*
2. What design principle does this demonstrate? *Name and explain the principle.*
3. What is the practical application in architecture? *Describe a real building example.*

Faculty signature: _____

SUPPORTING KNOWLEDGE**Design terminology and reference bank**

The official detailed syllabus focuses on visual understanding and representation. These reference entries support the official outcomes without increasing examination depth.

1. Scale

Scale = Drawing dimension : Real dimension = d/D

Symbols / units: d = measured dimension on drawing; D = actual dimension in same units. E.g. 1:100 means 1cm on drawing = 100cm (1m) in reality.

Example: A room 4m wide drawn at 1:50 scale: $400\text{cm} \div 50 = 8\text{cm}$ on paper.

2. Golden Ratio

$\phi = (1 + \sqrt{5}) / 2 \approx 1.618$

Symbols / units: ϕ (phi) is the golden ratio. If $a:b = (a+b):a$, the ratio is ϕ . Used in architectural proportions for aesthetic harmony.

Example: A wall 3.2m wide: divide at $3.2 \div 1.618 \approx 1.98\text{m}$ from one edge for golden section proportion.

3. Modulator

Modulator = Fibonacci $\times \phi$, human scale in cm

Symbols / units: Le Corbusier's system: a series of measurements derived from the human figure and golden ratio. Used to proportion architectural spaces to the human body.

Example: Standard Modulator dimensions: 113, 70, 43, 27, 16.5, 10 cm — use for furniture and room heights.

4. Value (Colour)

Value = Lightness or darkness of a colour on a scale 0-10

Symbols / units: 0 = absolute black, 10 = absolute white. Mid-value (5) is neutral grey. Higher value = lighter, lower = darker.

Example: To lighten a colour, add white (tint). To darken, add black (shade). A colour at value 7 appears lighter than the same colour at value 3.

5. Complementary Colours

Complementary pairs are opposite on the colour wheel

Symbols / units: Red↔Green, Blue↔Orange, Yellow↔Violet. When placed side by side they create maximum contrast and visual vibration.

Example: In a room: red walls with green accents create maximum visual contrast; use neutral to moderate.

6. Rhythm Types

5 types: Regular, Flowing, Progressive, Alternating, Radiating

Symbols / units: Regular = equal spacing; Flowing = curved progression; Progressive = graduated change; Alternating = two+ elements; Radiating = from centre.

Example: Colonnade with equal column spacing = regular rhythm; tapered columns = progressive rhythm.

7. Balance Types

Symmetrical, Asymmetrical, Radial

Symbols / units: Symmetrical = mirror image about axis; Asymmetrical = different elements balanced by visual weight; Radial = arranged around central point.

Example: Taj Mahal = symmetrical balance; Villa Savoye = asymmetrical balance; Pantheon = radial balance.

DESIGN PRACTICE

Worked design examples and practice

Practice Example 1: Determine the scale bar for a 1:50 drawing of a 6m × 4m room—

Given / observation: Room dimensions: 6m × 4m = 600cm × 400cm. Scale 1:50 means divide by 50.

Solution and conclusion: $600 \div 50 = 12\text{cm}$ and $400 \div 50 = 8\text{cm}$. Draw a rectangle 12cm × 8cm. Scale bar: 1cm on paper = 50cm real = mark every 1cm as 50cm.

Practice Example 2: Identify the colour scheme in a sunset photograph—

Given / observation: Sunset contains red, orange and yellow hues arranged adjacently on the colour wheel.

Solution and conclusion: These are analogous colours (adjacent on wheel). The scheme is analogous warm. Emotional effect: warmth, energy, drama.

Practice Example 3: Determine the dominant design principle in the Taj Mahal facade—

Given / observation: Taj Mahal has mirror-image left and right halves about a central vertical axis.

Solution and conclusion: The dominant principle is symmetrical balance. Sub-principles: hierarchical emphasis (central dome > minarets), proportional harmony (golden ratio in dome-to-base ratio).

Practice Example 4: Select appropriate line type for a roof slope in a sketch—

Given / observation: A roof slopes downward from ridge to eave, creating a diagonal line.

Solution and conclusion: Use a strong diagonal line to represent the roof slope. Diagonal lines create dynamism and tension — appropriate for a roof's structural energy. Label it as a structural line.

Practice Example 5: Choose texture for a hot-climate building facade—

Given / observation: Hot climate requires heat reflection and visual coolness.

Solution and conclusion: Use light-coloured smooth or semi-rough texture. Light colour reflects heat; smooth surface reduces heat absorption. Avoid dark rough textures which absorb heat.

Practice Example 6: Identify the Five Points of Architecture in Villa Savoye—

Given / observation: Le Corbusier defined five essential features of modern architecture.

Solution and conclusion: (1) Pilotis: columns lifting building above ground. (2) Free façade: non-load-bearing walls. (3) Open plan: no internal load-bearing walls. (4) Ribbon window: horizontal strip window. (5) Roof garden: flat roof as usable garden.

Practice Example 7: Calculate Modulator dimension for a study desk height—

Given / observation: Standard study desk height using Modulator: use the 70cm dimension.

Solution and conclusion: Modulator dimension 70cm is the standard desk height. This is derived from human elbow height when seated, making it ergonomically comfortable for writing and reading.

Practice**Example 8: Determine the appropriate shading technique for a rough stone wall—**

Given / observation: Rough stone has irregular, granular surface texture.

Solution and conclusion: Use cross-hatching (overlapping sets of parallel lines at different angles) to represent rough stone. Vary line density: denser hatching = darker shadow areas; lighter hatching = highlight areas. Add irregular marks to suggest stone edges.

DRAWING PRACTICE**Complete labelled diagram library****Architectural drawing sheet layout with title block**

Figure reference — reproduce in examination drawings

Figure 1. Architectural drawing sheet layout with title block. Reproduce the essential parts, labels and arrows in examination drawings.

Point line plane and volume elements

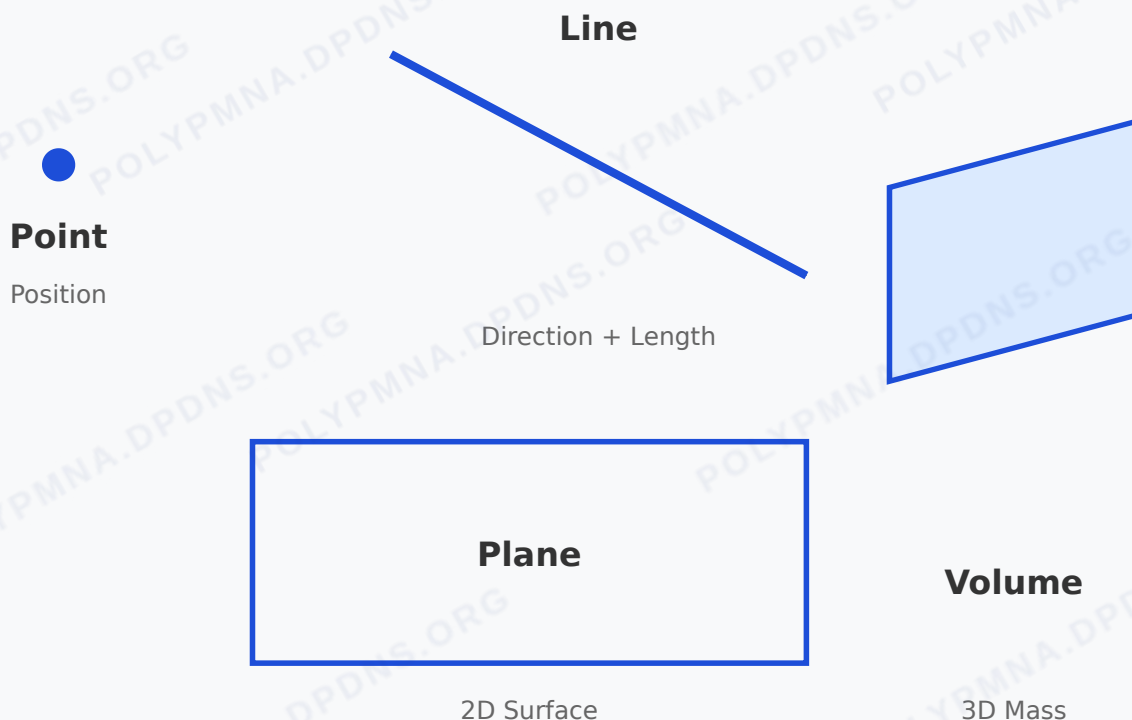


Figure 2. Point line plane and volume elements. Reproduce the essential parts, labels and arrows in examination drawings.

Line types: vertical horizontal diagonal zigzag curvilinear spiral

Figure reference — reproduce in examination drawings

Figure 3. Line types: vertical horizontal diagonal zigzag curvilinear spiral. Reproduce the essential parts, labels and arrows in examination drawings.

Solid and void composition with shading techniques

Figure reference — reproduce in examination drawings

Figure 4. Solid and void composition with shading techniques. Reproduce the essential parts, labels and arrows in examination drawings.

Colour wheel with primary secondary and tertiary colours

Figure reference — reproduce in examination drawings

Figure 5. Colour wheel with primary secondary and tertiary colours. Reproduce the essential parts, labels and arrows in examination drawings.

Texture types and architectural applications

Figure reference — reproduce in examination drawings

Figure 6. Texture types and architectural applications. Reproduce the essential parts, labels and arrows in examination drawings.

Principles of design: balance proportion rhythm hierarchy

Figure reference — reproduce in examination drawings

Figure 7. Principles of design: balance proportion rhythm hierarchy. Reproduce the essential parts, labels and arrows in examination drawings.

Freehand sketching process gesture study finished

Figure reference — reproduce in examination drawings

Figure 8. Freehand sketching process gesture study finished. Reproduce the essential parts, labels and arrows in examination drawings.

Kovalam Beach Resort architectural sketch

Figure reference — reproduce in examination drawings

Figure 9. Kovalam Beach Resort architectural sketch. Reproduce the essential parts, labels and arrows in examination drawings.

Sangath Doshi studio architectural sketch

Figure reference — reproduce in examination drawings

Figure 10. Sangath Doshi studio architectural sketch. Reproduce the essential parts, labels and arrows in examination drawings.

Rashtrapathi Bhavan architectural sketch

Figure reference — reproduce in examination drawings

Figure 11. Rashtrapathi Bhavan architectural sketch. Reproduce the essential parts, labels and arrows in examination drawings.

Falling Water Wright architectural sketch

Figure reference — reproduce in examination drawings

Figure 12. Falling Water Wright architectural sketch. Reproduce the essential parts, labels and arrows in examination drawings.

Villa Savoye Le Corbusier architectural sketch

Figure reference — reproduce in examination drawings

Figure 13. Villa Savoye Le Corbusier architectural sketch. Reproduce the essential parts, labels and arrows in examination drawings.

Farnsworth House Mies Van der Rohe architectural sketch

Figure reference — reproduce in examination drawings

Figure 14. Farnsworth House Mies Van der Rohe architectural sketch. Reproduce the essential parts, labels and arrows in examination drawings.

OFFICIAL 90-HOUR PLAN

Suggested student activities for self-learning

Week 1-2

Practice sheet layout, line weights and lettering daily. Create 3 compositions demonstrating vertical, horizontal and diagonal line types. Submit for review.

Week 3-4

Study colour wheel and mix primary, secondary, tertiary colours. Create monochromatic, analogous and complementary compositions. Identify colour schemes in daily life.

Week 5-6

Collect and mount 10 texture samples (5 natural, 5 artificial). Create drawn representations using cross-hatching, stippling and smudging. Label each sample.

Week 7-8

Develop 3 design composition sheets (balance, proportion/rhythm, hierarchy/contrast/harmony/emphasis). Annotate each principle. Revise based on faculty feedback.

Week 9-10

Visit 3 outdoor spaces. Produce gesture, study and finished sketches of each. Label elements and write design commentary. Focus on lighting and spatial quality.

Week 11-12

Study Indian buildings from reference images. Produce freehand sketches of Kovalam, Sangath and Rashtrapathi Bhavan. Write design commentary paragraphs.

Week 13-14

Study International buildings from reference images. Produce freehand sketches of Falling Water, Villa Savoye and Farnsworth House. Write philosophy commentary.

Week 15-16

Revise all modules. Complete practical record. Prepare for end-semester examination. Review expected questions and practice model paper.

EXAMINATION PREPARATION

Expected questions with answers

Questions follow the official 1-mark, 3-mark and 7-mark structure. Answers are original learning support, not copied model-paper text.

1-mark

What are the four primary elements of design?

Answer: Point, Line, Plane and Volume. Point has position only; line is a path of a point; plane is a two-dimensional surface; volume is a three-dimensional mass.

1-mark

Define scale in architectural drawing.

Answer: Scale is the ratio of the dimension on a drawing to the corresponding real dimension of the object or building. It is always shown as a scale bar.

1-mark

Name the three primary colours.

Answer: Red, Yellow and Blue.

1-mark

What is texture in design?

Answer: Texture is the perceived or actual surface quality of a material — how it looks and feels to touch.

1-mark

Name the three types of balance in design.

Answer: Symmetrical balance, Asymmetrical balance and Radial balance.

1-mark

What is the golden ratio?

Answer: The golden ratio (ϕ) is approximately 1:1.618, a mathematical proportion where the ratio of the whole to the larger part equals the ratio of the larger part to the smaller part.

1-mark

Who designed Kovalam Beach Resort?

Answer: Charles Correa.

1-mark

What is the Five Points of Architecture?

Answer: Pilotis, Free Façade, Open Plan, Ribbon Window and Roof Garden — defined by Le Corbusier for Villa Savoye.

1-mark

What does 'less is more' mean in architecture?

Answer: It is Mies Van der Rohe's principle of architectural minimalism — removing all non-essential elements to reveal the essential relationship between structure, space and nature.

1-mark

Name two shading techniques.

Answer: Hatching and cross-hatching. (Also: stippling, smudging)

3-mark

Explain the elements of design with their characteristics.

Answer: The four elements are: Point (position only, marks location), Line (path of a point, carries direction and character), Plane (two-dimensional surface, defines boundaries), Volume (three-dimensional mass, gives spatial presence). Each element builds upon the previous one.

3-mark

Describe the colour wheel and colour schemes.

Answer: The colour wheel organises hues by wavelength. Primary colours (red, yellow, blue) cannot be mixed. Secondary colours (orange, green, violet) come from mixing two primaries. Tertiary colours come from primary + adjacent secondary. Schemes: complementary (opposite), analogous (adjacent), triadic (evenly spaced), monochromatic (one hue).

3-mark

Explain the principles of balance and proportion.

Answer: Balance is the distribution of visual weight — symmetrical (mirror), asymmetrical (different elements balanced), radial (around centre). Proportion is the relationship between parts and the whole, and between the building and human body. The golden ratio (1:1.618) and Modulor are mathematical bases.

3-mark

Describe the architectural features of Falling Water.

Answer: Falling Water (Wright, 1935) is built over a waterfall in Pennsylvania. It features cantilevered concrete terraces extending over the stream, horizontal lines echoing rock strata, a natural rock hearth as the interior centre, and organic integration of rock and water. It exemplifies organic architecture.

3-mark

What are the Five Points of Architecture? Explain any three.

Answer: (1) Pilotis: reinforced concrete columns lift the building above ground. (2) Free Façade: walls are non-load-bearing, allowing windows of any shape. (3) Open Plan: no internal load-bearing walls, flexible interior. (4) Ribbon Window: horizontal strip window for uniform light. (5) Roof Garden: flat roof restores land taken by footprint.

3-mark

Compare hatching and cross-hatching techniques.

Answer: Hatching uses parallel lines at a consistent angle to create tone and shadow. Cross-hatching uses overlapping sets of parallel lines at different angles for darker, richer tones. Hatching is lighter and faster; cross-hatching produces deeper shadows and more texture. Both are used in architectural representation.

3-mark

Explain warm and cool colours with architectural application.

Answer: Warm colours (red, orange, yellow) appear to advance toward the viewer and make spaces feel intimate and energetic. Cool colours (blue, green, violet) appear to recede and make spaces feel

larger and calmer. In a small room, cool colours expand perceived space; in a large hall, warm colours create intimacy.

7-mark

Explain the elements of design and their application in architecture.

Answer: The four primary elements are Point, Line, Plane and Volume. A point has position only and marks a location — in architecture it becomes a column, lamp or threshold. A line is a moving point carrying direction and character — it becomes walls, cornices and railings. A plane is a two-dimensional surface defining boundaries — it becomes floors, walls and ceilings. Volume is three-dimensional mass giving spatial presence — it becomes rooms, halls and courtyards. Forms are classified as regular (geometric) or irregular (organic). The architect selects and combines these elements to create spatial narrative, rhythm and hierarchy.

7-mark

Describe colour theory and its application in architectural design.

Answer: Colour has three qualities: Hue (colour name), Value (lightness/darkness), Intensity (purity/saturation). Primary colours are red, yellow, blue. Secondary: orange, green, violet. Tertiary: from primary + adjacent secondary. The colour wheel organises these and identifies complementary (opposite), analogous (adjacent), triadic (evenly spaced) schemes. Colour psychology: red=energy, blue=calm, green=nature. Warm colours advance (intimate spaces); cool colours recede (expand space). In architecture, colour influences mood, scale perception, thermal comfort and cultural identity.

7-mark

Explain the principles of design with illustrations from famous buildings.

Answer: The principles are: Balance (Taj Mahal — symmetrical), Proportion and Scale (golden ratio 1:1.618), Rhythm (Parthenon colonnade — regular), Hierarchy (Falling Water — dominant cantilever), Contrast (Villa Savoye — white vs dark pilotis), Harmony (Sangath — warm sandstone palette), Emphasis (Rashtrapathi Bhavan — central dome). Each principle governs how elements are organised to create a successful composition.

7-mark

Describe the design features of Sangath (B.V. Doshi) and its architectural significance.

Answer: Sangath (1980, Ahmedabad) is Doshi's own studio. It integrates Indian vernacular architecture with modernist abstraction. Vaulted brick shells create cave-like interiors cooled by natural ventilation — no mechanical systems needed. Sunken gardens and water channels respond to the harsh Gujarat climate. The irregular plan encourages creative movement and spontaneous collaboration. The building is a living manifesto of sustainable, community-centred architecture. It proves that comfort can be achieved through ancient Indian passive cooling strategies reinterpreted through modern form.

7-mark

Describe the design features of Villa Savoye (Le Corbusier) and its architectural significance.

Answer: Villa Savoye (1929, Poissy, France) is the definitive manifesto of the International Style. It embodies Le Corbusier's Five Points: (1) Pilotis — reinforced concrete columns lift the white box above the green field; (2) Free Façade — non-load-bearing walls allow ribbon windows; (3) Open Plan — no internal load-bearing walls; (4) Ribbon Window — horizontal strip for uniform light; (5) Roof Garden — flat roof as usable garden. The pure white rendered surface and geometric purity express a machine-age aesthetic. It represents the universal, rational approach to modern architecture.

7-mark

Compare the architectural philosophies of Wright, Le Corbusier and Mies Van der Rohe.

Answer: Wright's organic architecture: building belongs to its site like a plant to soil — site-specific, natural materials, integration with landscape (Falling Water). Le Corbusier's International Style: machine-age rationalism — industrial materials, Modulor proportions, universal aesthetic transcending local tradition (Villa Savoye). Mies Van der Rohe's minimalism: 'less is more' — elimination of all non-essential elements to reveal structure-space-nature relationship (Farnsworth House). Wright = site-specific organic; Le Corbusier = universal machine; Mies = essential minimalism.

7-mark

Explain freehand sketching techniques and their role in architectural design.

Answer: Freehand sketching is the most direct means of architectural observation and communication. Three stages: Gesture drawing (30 seconds) captures overall impression and dominant lines. Study sketch (3-5 minutes) develops major shapes and spatial relationships. Finished sketch (15-30 minutes) applies lighting and shading for depth, form and spatial quality. The architect must identify and label elements, note light direction, and write commentary on design principles. This develops

hand-eye coordination, observational accuracy and analytical thinking — essential skills for architectural design and communication.

7-mark

Create a comprehensive study of texture in architectural design.

Answer: Texture is the surface quality of a material — tactile (felt) or visual (implied). Functions: visual interest, material honesty, acoustic properties (rough absorbs, smooth reflects), thermal performance (dark rough absorbs heat, light smooth reflects), cultural identity. Types: rough (stone, brick), smooth (marble, glass), glossy (granite), matte (concrete), patterned (tiles), organic (wood grain). In hot climates, use light smooth textures for heat reflection. In cold climates, dark rough textures absorb warmth. The architect selects texture to enhance design concept and respond to climate and culture.

OFFICIAL EVALUATION

Assessment pattern and practice model paper

Continuous Internal Evaluation — 40 marks

Component	Marks
Attendance	5
Continuous evaluation test	10
Series Test	10
Assignment / Practical	15

End-Semester Examination — 100 marks

Duration: 3 hours (including 15 min reading time)

Part	Description	Marks
Part A	2 short answer questions from each module	8
Part B	Answer 2 out of 3 questions from each module (3 marks each)	24
Part C	1 set of 7-mark questions with choice from each module	28

Total: 60 marks theory (practicum assessed separately)

Original practice model paper

Duration: 3 hours (including 15 min reading time) · **Maximum:** 60 marks

Part A — answer all ($8 \times 1 = 8$)

1. Define Basics of architectural drawing.
2. Define Elements of design: point, line, plane and volume.
3. Define Colour theory.
4. Define Texture in architectural design.
5. Define Principles of design.
6. Define Illustration of principles in architecture.
7. Define Introduction of famous Indian buildings.
8. Define Design features and significance of Indian buildings.

Part B — answer two from each module ($8 \times 3 = 24$)

Module 1: (1) Explain Basics of architectural drawing. (2) Explain Elements of design: point, line, plane and volume. (3) Explain Application of elements in architecture.

Module 2: (1) Explain Colour theory. (2) Explain Texture in architectural design. (3) Explain Colour scheme and the colour wheel.

Module 3: (1) Explain Principles of design. (2) Explain Illustration of principles in architecture. (3) Explain Representation of principles through design compositions.

Module 4: (1) Explain Introduction of famous Indian buildings. (2) Explain Design features and significance of Indian buildings. (3) Explain Freehand sketching of Indian buildings.

Part C — answer one alternative from each module ($4 \times 7 = 28$)

Module 1: (a) Explain Representation of elements: line types and visual impact with labelled figure and application. **OR** (b) Explain Representation: compositions of solids, voids, lighting and shading with principle, comparison and significance.

Module 2: (a) Explain Colour compositions: practical application with labelled figure and application. **OR** (b) Explain Representation of texture with principle, comparison and significance.

Module 3: (a) Explain Representation of principles through design compositions with labelled figure and application. **OR** (b) Explain Freehand sketching of outdoor spaces with principle, comparison and significance.

Module 4: (a) Explain Design features and significance of International buildings with labelled figure and application. **OR** (b) Explain Freehand sketching of International buildings with principle, comparison and significance.

LAST-MINUTE REVIEW

Quick revision and answer checklist

Module I anchors

Point, Line, Plane, Volume elements; architectural drawing sheet layout; line weight hierarchy; line types and visual impact; solid-void composition; shading techniques (hatching, cross-hatching, stippling,

smudging); scale definition.

Module II anchors

Colour qualities (hue, value, intensity); primary, secondary, tertiary colours; colour wheel and schemes (complementary, analogous, triadic, monochromatic); warm and cool colours; texture types (tactile, visual); texture functions; material mounting and drawn representation.

Module III anchors

Balance (symmetrical, asymmetrical, radial); proportion and scale (golden ratio 1:1.618, Modulor); rhythm types (regular, flowing, progressive, alternating, radiating); hierarchy; contrast; harmony; emphasis; freehand sketching stages (gesture, study, finished).

Module IV anchors

Kovalam Beach Resort (Correa): tropical modernism; Sangath (Doshi): vernacular + modernist; Rashtrapathi Bhavan (Lutyens): classical + Indian; Falling Water (Wright): organic architecture; Villa Savoye (Le Corbusier): Five Points, International Style; Farnsworth House (Mies): minimalism, 'less is more'.

Diagram checklist

Always include: title, correct outline, principal parts labelled, arrows for visual/spatial/light flow, input and output labels, scale bar where applicable, and a caption explaining the purpose.

Answer checklist

Start with definition, state principle or classification, list or compare components, use correct terminology, add labelled diagram or composition where applicable, state application or limitation, and finish with cultural or architectural significance.

OFFICIAL LEARNING RESOURCES

Textbooks, references and equipment

Official textbooks

1. *Graphic Thinking for Architects and Designers* — John Wiley & Sons, New York, 2001.
2. *Design Fundamentals in Architecture* — V.S. Pramar, Somaiya Publications Pvt. Ltd., New Delhi.
3. *Principles of Two-Dimensional Design* — Francis D.K. Ching, Van Nostrand Reinhold Co., Canada.

Official references

1. *The Fundamentals of Architecture* — Lorraine Farrelly, AVA Publishing, 2011.

Official web resources:

1. [BME Architecture Introduction — Construction PDF](#) (Modules I & II).
2. [ArchDaily](#) (Module IV).

Major equipment for 30 students

1. Drawing boards — 30.
2. Drawing sheets — 30.
3. Drawing tables — 30.
4. Drawing chairs — 30.
5. Drawing instruments: T-scale, set square — 30.

Official software requirement: NIL.

Studio-ൽ drawing instruments ശരിയായി maintain ചെയ്യുക; sharing പാടില്ല.

POLY PMNA · Course 1182 · Diploma Curriculum Revision 2026

Original bilingual educational handbook based on the supplied official syllabus. Always follow institution studio rules and faculty instructions.